1 Papers

1.1 Combining Diverse Anomaly Detectors for Automated Anomaly Labeling...

From [FO08]: unsupervised combination strategies. It is possible to simulate anomalies to evaluate the anomaly detector

2 Anomaly Detection, Principles and Algorithms

[ME17] (3): Anomaly detection= compare against expected or normal behavior or data. *Anomalies= substantial variations from the norm* (4). (4) The primary assumption of normal behavior is stationarity, so past statistics will be valid in the future. (6) In anomaly detection is not possible to use ML and DL, because anomalous cases are much rarer than non anomalous, moreover, anomalous cases may have little in common. (13)

2.1 Anomaly detection

Processes and behaviors result in the state of a system, manifested in observable data (21). From observable data we must formulate hypotheses about the underlying process, under the hipotheses that the system is in normal state, verified observing additional data.
Systems can also exist in abnormal states, leading to observable data . Anomaly detection tries to discover the variations from the norm. Three cases need to be considered (22): correct detection (detected anomalies correspond to actual anomalies in the process), false positives, false negatives. So the anomaly detection tast is estimating the likelihoot that a sample in anomalous (22).
(23) Metrics.

Precision	Recall	
$\frac{t_p}{t_p + f_p}$	$\frac{t_p}{t_p + f_n}$	

(24) "*observeble data do reflect anomalies in the underlying process or behavior*", but the changes in the data are not supposed to follow known patterns. The assumptions are (24):

- some process exists, that is not completaly random but that has rules;
- the characteristis of the process hold in the entire set of data. Samples with 10% of the data should have the same characteristics of the entire dataset.
- The process does not vary with time, but trends and cycles can exist.
  [26] If the distribution is normal, the Mahalanobis distance is the square root of the probability that a point belongs to the distribution (so, graph each feature to determine if its distribution is normal). If the distribution is not normal, the norm can be a set of points (26): an abnormal point is distant from all normal points.

[26] single dimensional data. In normal distributions, data beyond 3*sigma are anomalous. In caso of multimodal distributions, think data as a collection of clusters of data points. Clustering plays an important role in anomaly detection. Points distant from neighboring clusters are candidate to be anomalous [26]. Cluster: a group of that are near each other.
Near by density of by distance.

[28]

3 Biography

ID	Title	Version
[FO08]	"MAWILab : Combining Diverse Anomaly Detectors for Automated Anomaly Labeling and Performance Benchmarking", Fontugne et al	2008
[ME17]	"Anomaly Detection, Principles and Algorithms", K.G. Mehrotra, C.K. Mohan, H. Huang, Springer	1st edition, 2017